

QFT-22 By: C I P H E R

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1: (Only One Option Correct Type)

The standard deviations of two sets of observations $X = \{x_i\}$ and $Y = \{y_i\}$ ($i = 1, 2, \dots, 100$) are respectively 5 and 6. If $\bar{x}, \bar{y}$ are their means and $\sum_{i=1}^{100} (x_i - \bar{x})(y_i - \bar{y}) = 600$, then the standard deviation of $Z = \{z_i | z_i = x_i - y_i\}$ is

- (A) 6
- (B) 9
- (C) 8
- (D) 7

Question No. 2: (Only One Option Correct Type)

The sum of the infinitely decreasing geometric progression is equal to the greatest value of the function $f(x) = 3x^3 - x - 76$ on the interval $[0, 3]$; the first term of the progression is equal to the square of the common ratio. The common ratio of the G.P. is -

- (A) $\sqrt{2} - 1$
- (B) $\sqrt{3} - 1$
- (C) $\sqrt{2} + 1$
- (D) $\frac{1}{\sqrt{3}+1}$

Question No. 3: (Only One Option Correct Type)

Given real numbers x, y, z , such that $x + y + z = 3$, $x^2 + y^2 + z^2 = 5$, $x^3 + y^3 + z^3 = 7$, find $x^4 + y^4 + z^4$.

- (A) 18
- (B) 16
- (C) 9
- (D) 24

Question No. 4: (Only One Option Correct Type)

Let R be a relation defined on the set of natural numbers N as $R = \{(x, y) : x, y \in N, 2x + y = 41\}$. Which of the following is true?

- (A) R is reflexive
- (B) R is symmetric
- (C) R is transitive
- (D) At least one is false

Question No. 5: (Only One Option Correct Type)

If the line $y - \sqrt{3}x + 3 = 0$ cuts the parabola $y^2 = x + 2$ at A and B , if $PA \cdot PB = k \frac{(\sqrt{3}+2)}{3}$ [where P is $(\sqrt{3}, 0)$] then $k =$

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Question No. 6: (Only One Option Correct Type)

Three digit numbers in which the middle one is a perfect square are formed using the digits 1 to 9. Their sum is :

- (A) 139055
- (B) 136055
- (C) 132055
- (D) 134055

Question No. 7: (Only One Option Correct Type)

The perpendicular bisector of the line segment with endpoints $(2, 3, 2)$ and $(-4, 1, 4)$ passes through the point $(-3, 6, 1)$ and has equation of the form $\frac{x+3}{a} = \frac{y-6}{b} = \frac{z-1}{c}$ where a , b and c are relatively prime integers with $a > 0$. Find the value of $abc - (a + b + c)$:

- (A) -1
- (B) -2
- (C) -3
- (D) -4

Question No. 8: (Only One Option Correct Type)

Let $\vec{a} = 2\hat{i} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r} \cdot \vec{a} = 0$, then the value of $\vec{r} \cdot \vec{b}$ is

- (A) 7
- (B) -7
- (C) -5
- (D) 5

Question No. 9: (Only One Option Correct Type)

The value of $\int_{-(\pi/4)^{1/3}}^{(\pi/4)^{1/3}} \frac{x^2}{(1+\sin^2 x^3)(1+e^{x^7})} dx$ is equal to :

- (A) $\frac{1}{2\sqrt{3}} \tan^{-1} \sqrt{2}$
- (B) $\frac{1}{3\sqrt{2}} \tan^{-1} \sqrt{2}$
- (C) $\frac{1}{\sqrt{2}} \tan^{-1} \sqrt{2}$
- (D) None of these

Question No. 10: (Only One Option Correct Type)

The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets the auxiliary circle at the point M . The area of the triangle AMO is

- (A) $\frac{31}{10}$
- (B) $\frac{29}{10}$
- (C) $\frac{21}{10}$
- (D) $\frac{27}{10}$

Question No. 11: (Only One Option Correct Type)

If A, B are two non-singular matrices of order 3 and I is an identity matrix of order 3 such that $AA^T = 5I$ and $3A^{-1} = 2A^T - A \operatorname{adj}(4B)$, then $|B|^2$ is equal to (where A^T and $\operatorname{adj}(A)$ denote transpose and adjoint matrices of the matrix A respectively)

- (A) $\frac{7^3}{5^3 \cdot 4^6}$
- (B) $\frac{7^3}{5^3 \cdot 4^3}$
- (C) $\frac{7^4}{5^3 \cdot 2^{12}}$

(D) $\frac{5^6}{7^3 \cdot 2^{10}}$

Question No. 12: (Only One Option Correct Type)

If principal argument of z satisfying inequalities $|z - 3| \leq \sqrt{2}$ and $|z - 6 - 3i| \leq 2\sqrt{2}$ is θ , then $\tan 2\theta$ is equal to -

(A) $\frac{1}{4}$

(B) $\frac{8}{15}$

(C) $\frac{3}{4}$

(D) $\frac{7}{15}$

Question No. 13: (Only One Option Correct Type)

Let the equation of a line is $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{3}$. An insect starts flying from $P(1, 3, 2)$ in a straight line meeting the given line at a point $R(a, b, c)$ and then goes to the point $Q(6, 7, 5)$ in a straight line such that PR is perpendicular to RQ . Then, the least value of $7(a + b + c)$ is equal to

(A) 105

(B) 45

(C) 10

(D) 7

Question No. 14: (Only One Option Correct Type)

$$\text{Let } \Delta(x) = \begin{vmatrix} 3 + 2 \sin^4 x & 2 \cos^4 x & \sin^2 2x \\ 2 \sin^4 x & 3 + 2 \cos^4 x & \sin^2 2x \\ 2 \sin^4 x & 2 \cos^4 x & 3 + \sin^2 2x \end{vmatrix}$$

then $\int_{-\pi/2}^{\pi/2} x \Delta(x) dx$ equals

(A) 2

(B) 2.5

(C) 1.5

(D) 0

Question No. 15: (Only One Option Correct Type)

A circle with centre $(3\alpha, 3\beta)$ and of variable radius cuts the hyperbola $x^2 - y^2 = 36$ at the points P, Q, R and S . If the locus of centroid of $\triangle PQR$ is $(x - 2\alpha)^2 - (y - 2\beta)^2 = \lambda$, then the value of λ is

- (A) 3
- (B) -2
- (C) 4
- (D) -3

Question No. 16: (Only One Option Correct Type)

If, $\sum_{k=1}^n (k^2 + 5k + 7)(k + 2)! + 3.3! = 2003.2003!$ then the value of n is,

- (A) 1000
- (B) 2001
- (C) 2002
- (D) None of these

Question No. 17: (Only One Option Correct Type)

If 6th term in the expansion of $\left(\frac{3}{2} + \frac{x}{3}\right)^n$ is numerically greatest, when $x = 3$, then find the sum of possible integral values of n .

- (A) 23
- (B) 24
- (C) 25
- (D) 26

Question No. 18: (Only One Option Correct Type)

If y satisfies the differential equation $xydy = y(xy^3 + 1)dx$ & $y(1) = 2$, then mark incorrect option.

- (A) $\lim_{x \rightarrow \infty} xy^3 = -\frac{4}{3}$
- (B) $\lim_{x \rightarrow \infty} y = 0$
- (C) curve $y = f(x)$ is symmetric w.r.t. origin
- (D) curve $y = f(x)$ is continuous $\forall x \in \mathbb{R}$.

Question No. 19: (Only One Option Correct Type)

Equation of chord of a circle $S : (x - 3)^2 + (y - 4)^2 = 25$, which passes through two points on the circle situated at a distance 3 units from origin, is -

- (A) $2x + 3y - 3\sqrt{13} = 0$

(B) $6x + 8y = 9$

(C) $2x - 3y + 3\sqrt{13} = 0$

(D) $6x - 8y + 9 = 0$

Question No. 20: (Only One Option Correct Type)

The area of the region including between the regions satisfying $\min(|x|, |y|) \geq 1$ and $x^2 + y^2 \leq 5$ is

(A) $\frac{5}{2} \left(\sin^{-1} \frac{2}{\sqrt{5}} - \sin^{-1} \frac{1}{\sqrt{5}} \right) - 4$

(B) $10 \left(\sin^{-1} \frac{2}{\sqrt{5}} - \sin^{-1} \frac{1}{\sqrt{5}} \right) - 4$

(C) $\frac{2}{5} \left(\sin^{-1} \frac{2}{\sqrt{5}} - \sin^{-1} \frac{1}{\sqrt{5}} \right) - 4$

(D) $15 \left(\sin^{-1} \frac{2}{\sqrt{5}} - \sin^{-1} \frac{1}{\sqrt{5}} \right) - 4$

Mathematics Numerical (Maximum Marks: 20)

Question No. 21: (Numerical Type)

The least value of n for which $(n - 2)x^2 + 8x + (n + 4) > \sin^{-1}(\sin 12) + \cos^{-1}(\cos 12) \forall x \in R (n \in N)$ is

Question No. 22: (Numerical Type)

Let f be a derivable function satisfying the equation $\int_0^x f(t)dt + \int_0^x t \cdot f(x - t)dt = e^{-x} - 1$. $\int_0^1 f(x)dx$ is equal to $\frac{K}{e}$, find value of $K + 10$.

Question No. 23: (Numerical Type)

There are 7 distinguishable rings. The number of possible five-rings arrangements on the four fingers (except the thumb) of one hand (the order of the rings on each finger is to be counted and it is not required that each finger has a ring) is equal to

Question No. 24: (Numerical Type)

Let A and B be two non-singular matrices such that $A \neq I$, $B^3 = I$ and $AB = BA^2$, where I is the identity matrix, the least value of k such that $A^k = I$ is

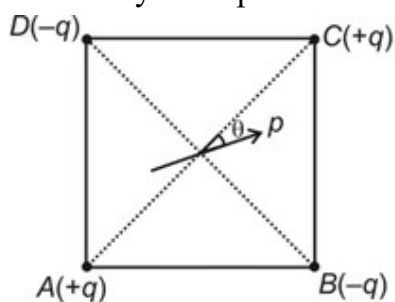
Question No. 25: (Numerical Type)

The value of $\lim_{n \rightarrow \infty} \left(\frac{[1^2x+1^2] + [2^2x+2^2] + \dots + [n^2x+n^2]}{n^3} \right)$ is $f(x)$, find $f(8)$. (where $[\cdot]$ denotes the greatest integer function.)

Physics Single Correct (Maximum Marks: 80)

Question No. 26: (Only One Option Correct Type)

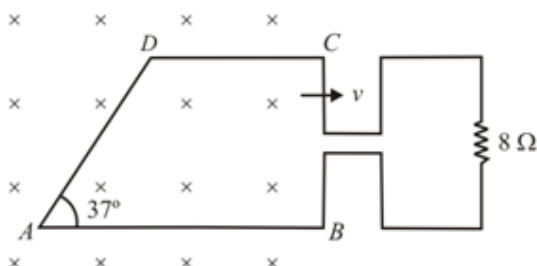
A square frame ABCD of side $r\sqrt{2}$ is made of insulated wires and there is a short dipole, with dipole moment p , fixed in the plane of the figure. The dipole is lying at the centre of the square, making an angle θ , as shown in the figure. If four point charges are placed at the four corners of the square, then the magnitude of force exerted by the dipole on the system of charges is



- (A) $\frac{6Kpq}{r^3}$
- (B) $\frac{4Kpq}{r^3}$
- (C) $\frac{2Kpq}{r^3}$
- (D) $\frac{8Kpq}{r^3}$

Question No. 27: (Only One Option Correct Type)

The loop ABCD is moving with velocity v towards right. The magnetic field is 4 T. The loop is connected to a resistance of 8Ω . If steady current of 2 A flows in the loop, then value of v , if loop has resistance of 4Ω , is (given, $AB = 30 \text{ cm}$, $AD = 30 \text{ cm}$)

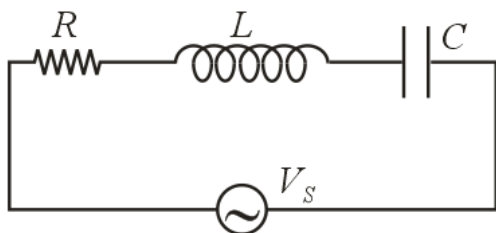


- (A) 20 m s^{-1}

- (B) 10 m s^{-1}
- (C) $\frac{100}{3} \text{ m s}^{-1}$
- (D) $\frac{10}{3} \text{ m s}^{-1}$

Question No. 28: (Only One Option Correct Type)

Assertion : In a series R , L , C circuit if V_R , V_L , and V_C denote *r m s* voltage across R , L and C respectively and V_s is the rms voltage across the source, then $V_s = V_R + V_L + V_C$.



Reason: Kirchoff voltage law is valid for DC as well as for AC circuit.

- (A) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- (B) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
- (C) Assertion is true but Reason is false.
- (D) Assertion is false but Reason is true.

Question No. 29: (Only One Option Correct Type)

Consider elastic collision of a particle of mass m moving with a velocity u with another particle of the same mass at rest. After the collision the projectile and the struck particle move in directions making angles θ_1 and θ_2 respectively with the initial direction of motion. The sum of the angles $\theta_1 + \theta_2$, is :

- (A) 45°
- (B) 90°
- (C) 135°
- (D) 180°

Question No. 30: (Only One Option Correct Type)

Assertion: A changing electric field produces a magnetic field.

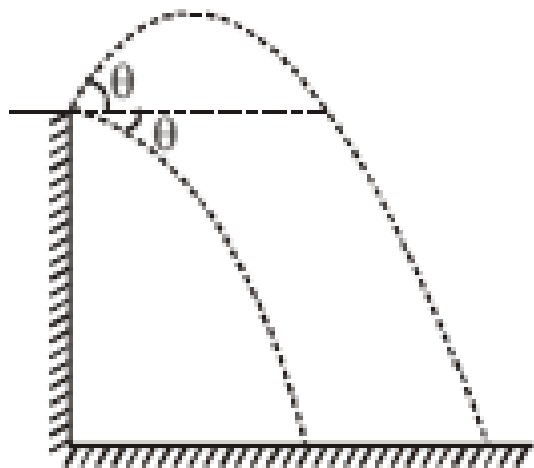
Reason: A changing magnetic field produces an electric field.

- (A) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

- (B) If both Assertion and Reason are true but Reason is not explanation of the Assertion.
- (C) If Assertion is true but the Reason is false.
- (D) If Assertion is false but Reason is true.

Question No. 31: (Only One Option Correct Type)

Two balls are thrown with the same speed v_0 from the top of a cliff. The angles of their initial velocities are θ above and below the horizontal, as shown. How much farther along the ground does the top ball hit than the bottom ball :-



- (A) $\frac{2v_0^2 \sin^2 \theta \cos^2 \theta}{g}$
- (B) $\frac{2v_0^2 \sin \theta}{g}$
- (C) $\frac{2v_0^2 \cos \theta}{g}$
- (D) $\frac{2v_0^2 \sin \theta \cos \theta}{g}$

Question No. 32: (Only One Option Correct Type)

The potential energy of particle of mass m varies as $U(x) = \begin{cases} E_0 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{for } x > 1 \end{cases}$ The de Broglie wavelength of the particle in the range $0 \leq x \leq 1$ is λ_1 and that in the range $x > 1$ is λ_2 . If the total energy of the particle is $2E_0$, find λ_1/λ_2

- (A) $\sqrt{2}$
- (B) $\sqrt{3}$
- (C) $\sqrt{\frac{1}{2}}$
- (D) $\sqrt{\frac{2}{3}}$

Question No. 33: (Only One Option Correct Type)

Given below are two statements

Statement I: In hydrogen atom, the frequency of radiation emitted when an electron jumps from lower energy orbit (E_1) to higher energy orbit (E_2), is given as $hf = E_1 - E_2$

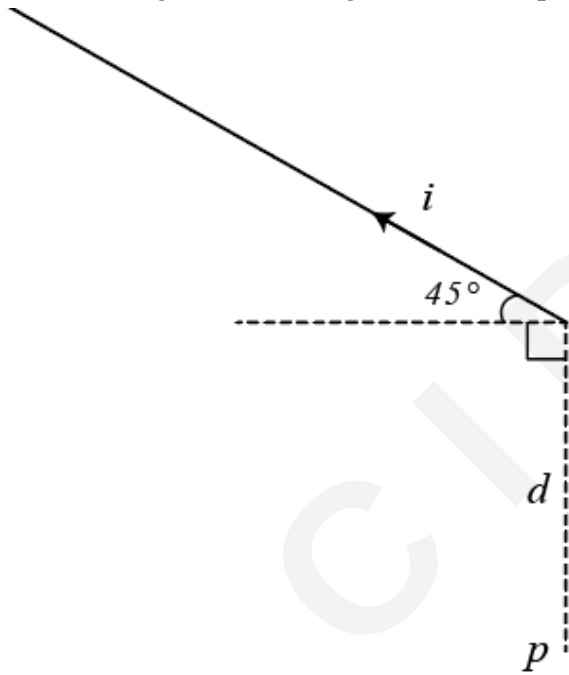
Statement II: The jumping of electron from higher energy orbit (E_2) to lower energy orbit (E_1) is associated with frequency of radiation given as $f = \frac{(E_2 - E_1)}{h}$. This condition is Bohr's frequency condition.

In the light of the above statements, choose the correct answer from the options given below:

- (A) Both statement I and statement II are true.
- (B) Both statement I and statement II are false.
- (C) Statement I is correct but statement II is false.
- (D) Statement I is incorrect but statement II is true.

Question No. 34: (Only One Option Correct Type)

Find the magnitude of magnetic field at point p due to a semi - infinite wire given below :-

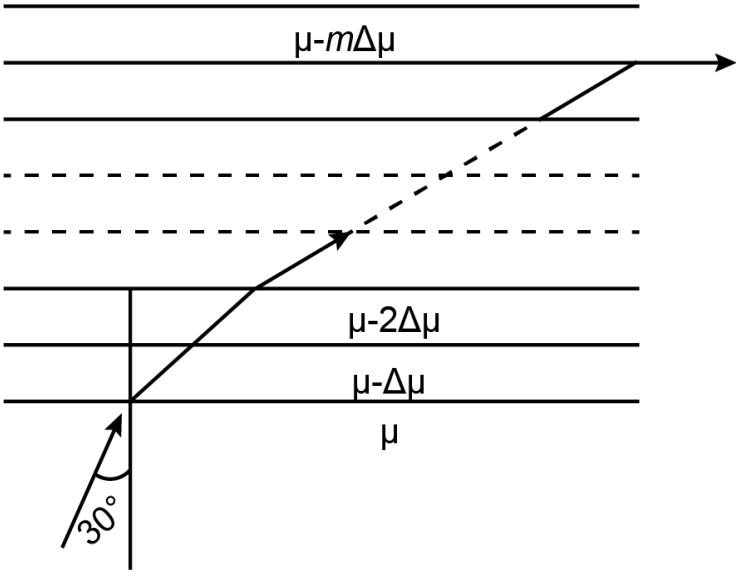


- (A) $\frac{\mu_0 i}{4\pi d}(\sqrt{2} - 1)$
- (B) $\frac{\mu_0 i}{4\pi d}(1 - \sqrt{2})$
- (C) $\frac{\mu_0 i}{4\pi d}\left(1 - \frac{1}{\sqrt{2}}\right)$
- (D) $\frac{\sqrt{2}\mu_0 i}{4\pi d}$

Question No. 35: (Only One Option Correct Type)

A glass slab consists of thin uniform layers of progressively decreasing refractive indices RI (see figure) such that the RI of any layer is $\mu - m\Delta\mu$. Here, μ and $\Delta\mu$ denote the RI of 0 th layer and the difference in RI between any two consecutive layers, respectively. The integer $m = 0, 1, 2, 3, \dots$ denotes the numbers of the

successive layers. A ray of light from the oth layer enters the 1st layer at an angle of incidence of 30° . After undergoing the mth refraction, the ray emerges parallel to the interface. If $\mu = 1.5$ and $\Delta\mu = 0.015$, the value of m is



- (A) 20
- (B) 30
- (C) 40
- (D) 50

Question No. 36: (Only One Option Correct Type)

A point object is moving at a constant speed of 1 ms^{-1} along the principal axis of a convex lens of focal length 10 cm. The speed of the image is also 1 ms^{-1} , when the object is at cm from the optical centre of the lens.

- (A) 10
- (B) 15
- (C) 20
- (D) 5

Question No. 37: (Only One Option Correct Type)

Column I specifies a point at distance r from the centre/axis of a symmetrical distribution of charge. Column II gives the variation of electric intensity at P as a function of r . Match the columns.

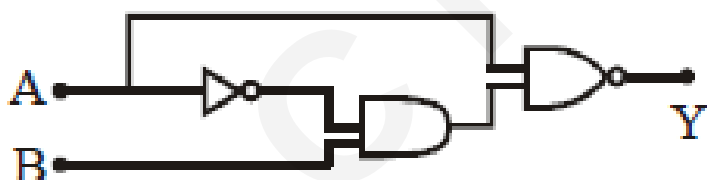
Column I	Column II
i. P lies outside a cylinder having uniform volume charge density.	a. $E \propto \frac{1}{r^2}$

Column I	Column II
ii. P lies inside a spherical charged conductor.	b. $E \propto \frac{1}{r}$
iii. P lies inside a spherical body having uniform volume charge density.	c. $E \propto r$
iv. P lies inside a solid cylinder having uniform volume charge density.	d. $E \propto r^0$
v. P lies inside a plane infinite sheet with some thickness. The sheet is charged uniformly throughout its volume.	
vi. P lies outside the plane sheet mentioned in part (v).	

- (A) i. $\rightarrow$ b., ii. $\rightarrow$ d., iii. $\rightarrow$ c. iv. $\rightarrow$ c., v. $\rightarrow$ c., vi. $\rightarrow$ d.
- (B) i. $\rightarrow$ a., ii. $\rightarrow$ d., iii. $\rightarrow$ c. iv. $\rightarrow$ b., v. $\rightarrow$ c., vi. $\rightarrow$ d.
- (C) i. $\rightarrow$ d., ii. $\rightarrow$ a., iii. $\rightarrow$ b. iv. $\rightarrow$ b., v. $\rightarrow$ d., iii. $\rightarrow$ a.
- (D) i. $\rightarrow$ c., ii. $\rightarrow$ d., iii. $\rightarrow$ a. iv. $\rightarrow$ b., v. $\rightarrow$ a., iii. $\rightarrow$ d.

Question No. 38: (Only One Option Correct Type)

The circuit is equivalent to



- (A) AND gate
- (B) OR gate
- (C) NOT gate
- (D) NOR Gate

Question No. 39: (Only One Option Correct Type)

The time dependence of a physical quantity p is given by $p = p_0 \exp(-\alpha t^2)$ where α is a constant and t is the time. The constant α

- (A) has dimensions of p

- (B) has dimensions of T^2
- (C) has dimensions of T^{-2}
- (D) is dimensionless

Question No. 40: (Only One Option Correct Type)

Match the following

	Column I		Column II
(a)	Enthalpy	(p)	Intensive property
(b)	Temperature	(q)	Path function
(c)	Free energy	(r)	Function of U, P and V
(d)	Work	(s)	State function

- (A) $a \rightarrow r, s, b \rightarrow p, s, c \rightarrow r, s, d \rightarrow q$
- (B) $a \rightarrow p, b \rightarrow r, c \rightarrow q, d \rightarrow r$
- (C) $a \rightarrow r, b \rightarrow q, c \rightarrow r, d \rightarrow p$
- (D) $a \rightarrow q, b \rightarrow r, c \rightarrow p, d \rightarrow r$

Question No. 41: (Only One Option Correct Type)

A slab of side 50 cm and thickness 10 cm is subjected to a shearing force of 10^5 N on its narrow edge. If the lower edge is riveted to the floor and upper edge is displaced by 0.2 mm, then shear modulus of the material of the slab is _____ GPa.

- (A) 4
- (B) 4.5
- (C) 5
- (D) 5.5

Question No. 42: (Only One Option Correct Type)

The speed of light (c), gravitational constant (G) and Planck's constant (h) are taken as fundamental units in a system. The dimensions of time in this new system should be:

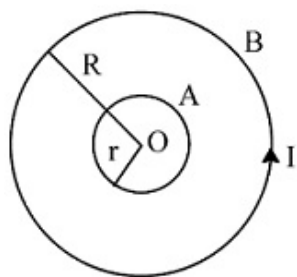
- (A) $G^{\frac{1}{2}} h^{\frac{1}{2}} c^{-\frac{5}{2}}$
- (B) $G^{-\frac{1}{2}} h^{\frac{1}{2}} c^{\frac{1}{2}}$

(C) $G^{\frac{1}{2}} h^{\frac{1}{2}} c^{-\frac{3}{2}}$

(D) $G^{\frac{1}{2}} h^{\frac{1}{2}} c^{\frac{1}{2}}$

Question No. 43: (Only One Option Correct Type)

O is the centre of two coplanar concentric circular conductors, A and B, of radii r and R respectively as shown in the fig. Here. $r \ll R$. The mutual inductance of the system of the conductors can be given by



(A) $\frac{\mu_0 \pi r}{2R}$

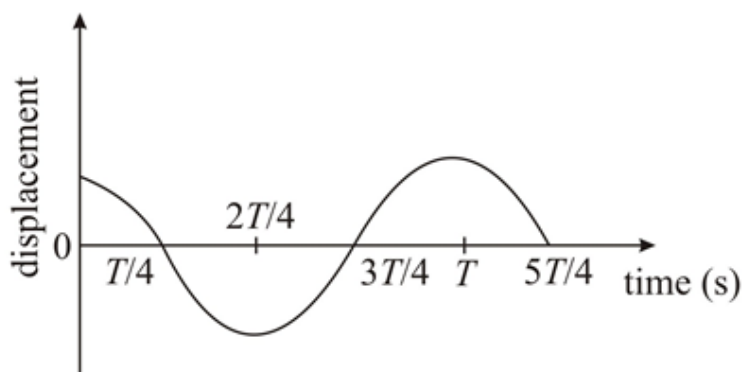
(B) $\frac{\mu_0 R^2}{\pi_0 r}$

(C) $\frac{\mu_0 \pi R^2}{2r}$

(D) $\frac{\mu_0 \pi r^2}{2R}$

Question No. 44: (Only One Option Correct Type)

The displacement time graph of a particle executing SHM is given in figure: (sketch is schematic and not to scale)



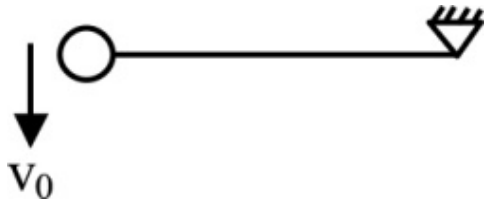
Which of the following statements is/are true for this motion?

- (A) The force is zero at $t = \frac{3T}{4}$
 (B) The magnitude of acceleration is maximum at $t = T$
 (C) The speed is maximum at $t = \frac{T}{4}$
 (D) The $P.E.$ is equal to $K.E.$ of the oscillation at $t = \frac{T}{2}$

- (A) (A), (B) and (C)
 (B) (B), (C) and (D)
 (C) (A), (B) and (D)
 (D) (A) and (D)

Question No. 45: (Only One Option Correct Type)

A small sphere is given vertical velocity of magnitude $v_0 = 5 \text{ m s}^{-1}$, and it swings in a vertical plane about the end of a massless string of length 2m. If maximum tension string can withstand is equal to twice the weight of sphere, then angle θ with vertical at which string will break is



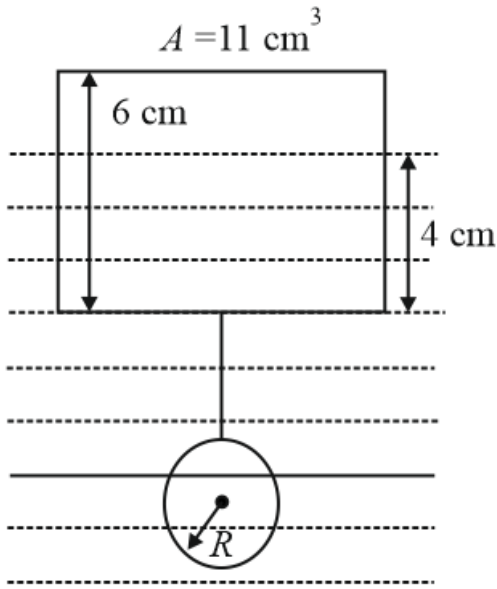
- (A) $\cos^{-1}\left(\frac{2}{3}\right)$
 (B) $\cos^{-1}\left(\frac{1}{4}\right)$
 (C) 60°
 (D) 30°

Physics Numerical (Maximum Marks: 20)

Question No. 46: (Numerical Type)

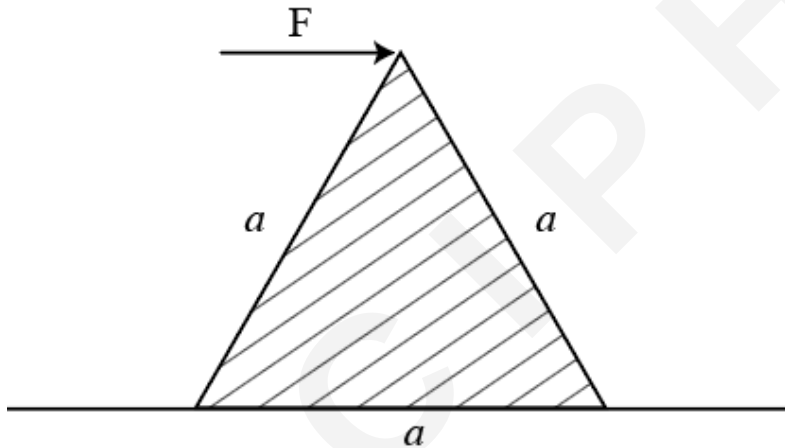
Figure shows a uniform metal ball suspended by thread of negligible mass from an upright cylinder that floats partially submerged in water. The cylinder has height 6 cm, face area 11 cm^2 on the top and bottom and density 0.5 g cm^{-3} , 4 cm of cylinder's height is inside the water surface. Density of the metal ball is 8 g cm^{-3} , R is the radius of the ball. It is found that $R^3 = \frac{3}{\alpha} \text{ cm}^3$, where α is an integer. Find

$\alpha. (\rho_w = 1 \text{ g cm}^{-3})$



Question No. 47: (Numerical Type)

An equilateral prism of mass m rests on a rough horizontal surface with coefficient of friction μ . A horizontal force F is applied on the prism as shown in the figure. If the coefficient of friction is sufficiently high so that the prism does not slide before toppling, then the minimum force required to topple the prism is $\frac{mg}{\sqrt{N}}$, Find N .



Question No. 48: (Numerical Type)

A thin walled glass sphere of radius 2.5 cm is filled with water. An object (O) is placed at 7.5 cm from the surface of the sphere. Neglecting the effect of glass wall, at what distance (in cm) the final image (I) of the object, measured from the centre of sphere is formed? (refractive index of water is $4/3$)

Question No. 49: (Numerical Type)

In an adiabatic expansion, the volume of air is increased by 5%, then the approximate percentage decrease in its pressure will be (Nearest Integer)

Question No. 50: (Numerical Type)

Consider a vehicle moving with a velocity 54 km/h. At a distance of 400 m, from the traffic light brakes are

applied. The acceleration of the vehicle, after the application of brakes is -0.3 m/s^2 . The vehicle's position relative to the traffic light is (till it stops)

Chemistry Single Correct (Maximum Marks: 80)

Question No. 51: (Only One Option Correct Type)

Match the complex given in column-I with their hybridisation given in column-II.

Column-I	Column-II
A. $[\text{Ni}(\text{CO})_4]$	1. dsp^2
B. $[\text{Ni}(\text{CN})_4]^{2-}$	2. sp^3
C. $[\text{Ni}(\text{NH}_3)_6]^{2+}$	3. d^2sp^3
D. $[\text{Fe}(\text{CN})_6]^{3-}$	4. sp^3d^2

(A)

A	B	C	D
1	2	3	4

(B)

A	B	C	D
2	1	4	3

(C)

A	B	C	D
1	3	2	4

(D)

A	B	C	D
2	3	1	4

Question No. 52: (Only One Option Correct Type)

The combustion of benzene (l) gives $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{l})$. Given that heat of combustion of benzene at constant volume is $-3263.9 \text{ kJmol}^{-1}$ heat of combustion (in kJmol^{-1}) of benzene at constant pressure will be ($R = 8.314 \text{ JK}^{-1} \text{ mol}^{-1}$) (given $T = 298 \text{ K}$)

(A) -3267.6

(B) 4152.6

(C) -452.46

(D) 3260

Question No. 53: (Only One Option Correct Type)

The dissociation equilibrium of a gas AB_2 can be represented as, $2\text{AB}_2(\text{g}) \rightleftharpoons 2\text{AB}(\text{g}) + \text{B}_2(\text{g})$. The degree

of dissociation is 'x' and is small compared to 1. The expression relating the degree of dissociation (x) with equilibrium constant K_P and total pressure P is

- (A) $\left(\frac{K_P}{P}\right)$
 (B) $\left(\frac{2K_P}{P}\right)$
 (C) $\left(\frac{2K_P}{P}\right)^{1/3}$
 (D) $\left(\frac{2K_P}{P}\right)^{1/2}$

Question No. 54: (Only One Option Correct Type)

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) :

Ionisation enthalpy increases along each series of the transition elements from left to right. However, small variations occur.

Reason (R) :

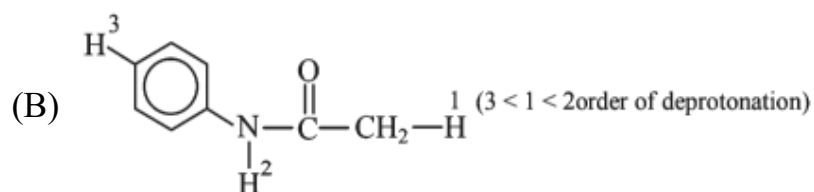
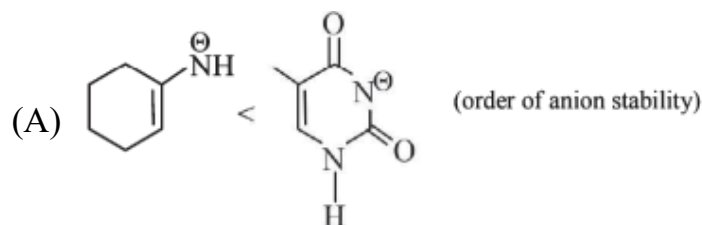
There is corresponding increase in effective nuclear charge which accompanies the filling of electrons in the inner d-orbitals.

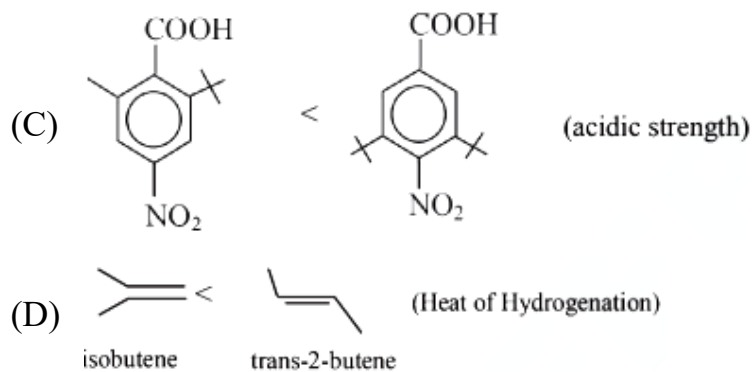
In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (B) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (C) (A) is correct but (R) is not correct
 (D) (A) is not correct but (R) is correct

Question No. 55: (Only One Option Correct Type)

Identify the incorrect option according to the property given in the parenthesis-





Question No. 56: (Only One Option Correct Type)

Identify the correct statements from the following

- (i) Electromeric effect is a permanent effect.
- (ii) Hyper conjugation is a temporary effect.
- (iii) Fractional distillation is used to separate two liquids from a mixture if the difference in their boiling points is less.
- (iv) Different compounds are adsorbed on an adsorbent different extents.

- (A) ii, iii, iv
- (B) i, ii, iii
- (C) ii, iv
- (D) iii, iv

Question No. 57: (Only One Option Correct Type)

Which of the following hydrides of nitrogen is acidic in nature?

- (A) NH_3
- (B) N_3H
- (C) N_2H_2
- (D) N_2H_4

Question No. 58: (Only One Option Correct Type)

Match List-I with List-II.

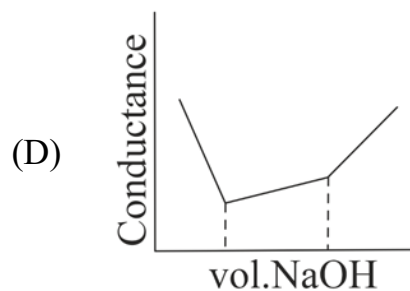
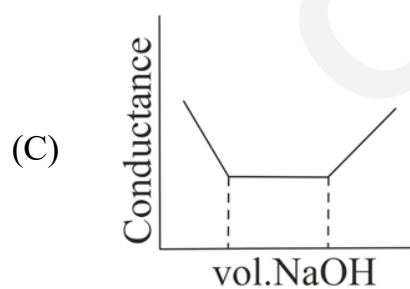
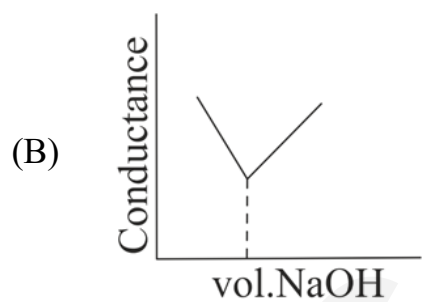
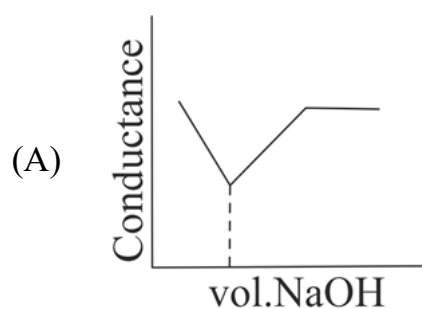
List-I		List-II	
(a)	Sucrose	(i)	β – D–Galactose and β – D–Glucose
(b)	Lactose	(ii)	α – D–Glucose and β – D–Fructose
(c)	Maltose	(iii)	α – D–Glucose and α – D–Glucose

Choose the correct answer from the options given below:

- (A) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (i)
(B) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (ii)
(C) (a) $\rightarrow$ (ii), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iii)
(D) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (ii)

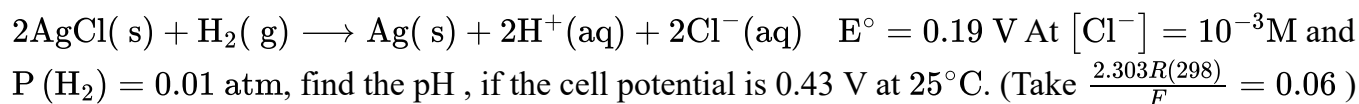
Question No. 59: (Only One Option Correct Type)

Among the following plots, which plot will obtain for a conductometric titration of a equimolar mixture of HCl and HCN with $\text{NaOH}_{(\text{aq})}$ solution?



Question No. 60: (Only One Option Correct Type)

Consider the following cell reaction :



- (A) 2
- (B) 3
- (C) 4
- (D) 1

Question No. 61: (Only One Option Correct Type)

Which one of the following on treatment with 50% aqueous sodium hydroxide yields the corresponding alcohol and acid?

- (A) $\text{C}_6\text{H}_5\text{CHO}$
- (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
- (C) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$
- (D) $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$

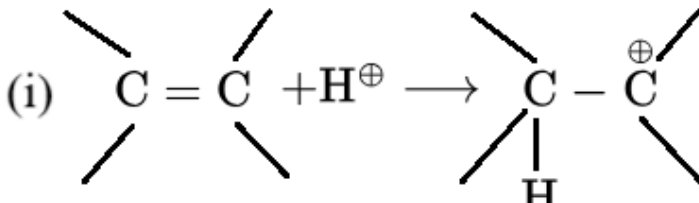
Question No. 62: (Only One Option Correct Type)

In an atom, an electron is moving with a speed of 600 m/s with an accuracy of 0.005%. Certainty with which the position of the electron can be located is ($h = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$, mass of electron, $m_e = 9.1 \times 10^{-31} \text{ kg}$)

- (A) $1.52 \times 10^{-4} \text{ m}$
- (B) $5.10 \times 10^{-3} \text{ m}$
- (C) $1.92 \times 10^{-3} \text{ m}$
- (D) $3.84 \times 10^{-3} \text{ m}$

Question No. 63: (Only One Option Correct Type)

Match the following.

List I	List II
(A) Resonance	(i) 
(B) Inductive effect	(ii) $\text{H} - \text{CH}_2 - \overset{\oplus}{\text{C}}\text{H}_2 \longleftrightarrow \overset{\oplus}{\text{H}} - \text{CH}_2 = \text{CH}_2$
(C) Electromeric effect	(iii) C_6H_6
(D) Hyperconjugation	(iv) $\text{CH}_3 - \text{Z} \longrightarrow \overset{\ominus}{\text{C}}\text{H}_3 + \overset{\oplus}{\text{Z}}$
	(v) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2\text{Cl}$

The correct answer is

A B C D

(A) II I IV III

(B) III V I II

(C) I III II V

(D) III II I IV

Question No. 64: (Only One Option Correct Type)

A hypothetical reaction, $A_2 + B_2 \longrightarrow 2AB$, follows the mechanism:

$A_2 \rightleftharpoons A + A$ (fast)

$A + B_2 \longrightarrow AB + B$ (slow)

$A + B \longrightarrow AB$ (fast)

The order of the overall reaction is

(A) 2

(B) 1

(C) 1.5

(D) zero

Question No. 65: (Only One Option Correct Type)

Which one of the following solutions in water will have the highest freezing point?

- (A) Solution which is 0.1 M in KNO_3 and 0.2 M in $\text{Ba}(\text{NO}_3)_2$.
- (B) Solution which is 0.1 M in $\text{Ca}(\text{NO}_3)_2$ and 0.1M $\text{Ba}(\text{NO}_3)_2$.
- (C) Solution which is 0.2 M in urea and 0.2 M glucose.
- (D) Solution which is 0.1 M both in $\text{Al}_2(\text{SO}_4)_3$ and 0.1M K_2SO_4 .

Question No. 66: (Only One Option Correct Type)

Study the following table.

Compound (Molecular weight)	Weight of compound (taken in g)
(a) CO_2 (44)	4.4
(b) NO_2 (46)	2.3
(c) H_2O_2 (34)	6.8
(d) SO_2 (64)	1.6

Which of these two compounds have the least weight of oxygen in them?

- (A) b and d
- (B) a and c
- (C) a and b
- (D) c and d

Question No. 67: (Only One Option Correct Type)

Among NF_3 and NCl_3 which statement is not correct?

- (A) Melting point of NF_3 is lower than NCl_3 .
- (B) Bond angle of $\text{F} - \text{N} - \text{F}$ is greater than $\text{Cl} - \text{N} - \text{Cl}$.
- (C) The bond length of $\text{N} - \text{F}$ is shorter than $\text{N} - \text{Cl}$.
- (D) Dipole moment of NCl_3 is greater than NF_3 .

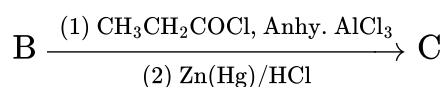
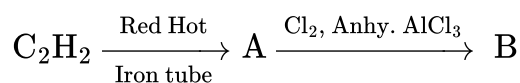
Question No. 68: (Only One Option Correct Type)

At 298 K, the standard electrode potentials of $\text{Cu}^{2+} / \text{Cu}$, $\text{Zn}^{2+} / \text{Zn}$, $\text{Fe}^{2+} / \text{Fe}$ and Ag^+ / Ag are 0.34 V, -0.76 V, -0.44 V and 0.80 V, respectively.

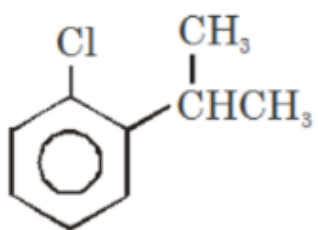
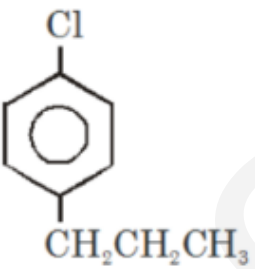
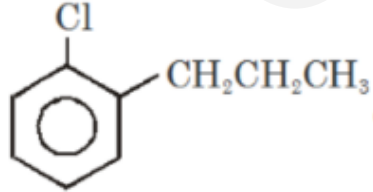
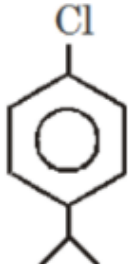
On the basis of standard electrode potential, predict which of the following reaction cannot occur?

- (A) $\text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$
- (B) $\text{FeSO}_4(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Fe}(\text{s})$
- (C) $2 \text{CuSO}_4(\text{aq}) + 2 \text{Ag}(\text{s}) \rightarrow 2 \text{Cu}(\text{s}) + \text{Ag}_2\text{SO}_4(\text{aq})$
- (D) $\text{CuSO}_4(\text{aq}) + \text{Zn}(\text{s}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s})$

Question No. 69: (Only One Option Correct Type)

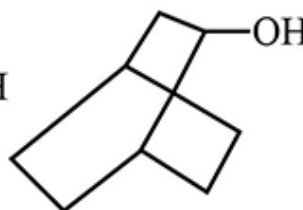
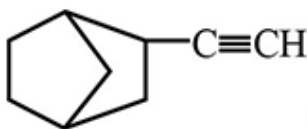
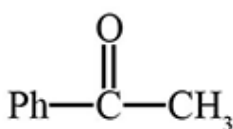
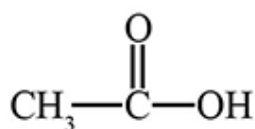
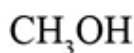
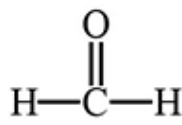
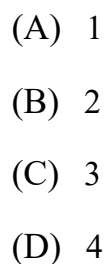


The major product 'C' is

- (A) 
- (B) 
- (C) 
- (D) 

Question No. 70: (Only One Option Correct Type)

Among the given complexes how many possess "CO" ligand bridges are



(VII)

Question No. 75: (Numerical Type)

How many grams of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, should be dissolved in 0.5 kg of water at 25°C to reduce the vapour pressure of water by 1%? (mark 10 times of mass of glucose as answer)

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